

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I G Amrutha(192511250) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: amrutha

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

AI-Powered Drone for Emergency Medicine Delivery Scenario

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is dynamic because roads may become blocked, weather conditions change travel costs, and the drone receives only partial real-time information. The drone must quickly find a safe and efficient route.

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the next node based only on the heuristic value $h(n)$, which estimates the distance from the current location to the goal. In this scenario, the heuristic is the straight-line distance to the destination.

How it behaves

- The drone always moves toward the location that appears closest to the destination.
- It ignores the actual travel cost caused by weather or terrain.
- If a road becomes blocked, the drone must search for another route, which may increase delay.

Strengths

- Fast decision-making.
- Requires less memory.
- Suitable for emergency situations where a quick route is needed.

Weaknesses

- Does not guarantee the shortest or safest path.
- Ignores travel cost and risk.
- May choose blocked or dangerous routes because it only considers the heuristic.

Conclusion:

Greedy Best-First Search is useful for fast decisions but is less reliable in uncertain and dynamic environments.

ii. Explain how A would handle the same situation, including how it balances cost and heuristic*

A Search*

A* Search evaluates each node using the function:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = Actual travel cost from the start to the current node.
- $h(n)$ = Estimated distance from the current node to the goal.

How A behaves*

- The drone considers both the actual travel cost and the estimated remaining distance.
- It avoids expensive routes caused by bad weather or rough terrain.
- When blocked roads are detected, A* recalculates and finds a better alternative path.

Advantages

- Finds the optimal path when the heuristic is accurate.
- Balances speed and path quality.
- Produces safer and more reliable routes.

Disadvantages

- Uses more memory.
- Requires more computation than Greedy Search.

Conclusion:

A* is better suited for emergency drone navigation because it considers both travel cost and estimated distance.

iii. Analyze the impact of heuristic accuracy on both algorithms

The quality of the heuristic greatly affects the performance of both search algorithms.

Heuristic Accuracy	Greedy Search	Best-First A Search*
High Accuracy	Quickly finds a good path	Quickly finds the optimal path
Moderate Accuracy	May choose inefficient routes	Still performs well because actual cost is considered
Poor Accuracy	Easily follows wrong directions	Explores more nodes but usually still finds a good solution

Analysis

- Greedy Search depends entirely on the heuristic.
- A* is more robust because it combines heuristic information with actual travel cost.

iv. Discuss how dynamic changes (blocked paths) affect search performance

Greedy Best-First Search

- May continue toward blocked roads.
- Often needs to restart the search.
- Search efficiency decreases significantly.

A Search*

- Updates the route when new obstacles are detected.
- Finds alternative paths using both cost and heuristic.
- Handles changing environments more effectively.

Impact

Dynamic obstacles increase the search complexity for both algorithms, but A* adapts more efficiently and produces safer routes.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

Recommended Algorithm: A* Search

Justification

- Considers both actual travel cost and estimated distance.
- Finds optimal and safer routes.
- Adapts better to blocked roads and changing weather conditions.
- Reduces the risk of delays in emergency medicine delivery.
- Although it requires more computation than Greedy Search, it provides higher reliability and safety.

Final Conclusion

For emergency medicine delivery in a flood-affected region, *A Search** is the most suitable algorithm because it balances **real-time decision-making, optimality, and safety**. Greedy Best-First Search is faster but may produce inefficient or unsafe routes due to its reliance only on heuristic information.